

0.1 Abel's Formular and Abel's Test

0.1.1 Abel's Formular: Summation by Parts

Lemma 0.1.1. Given sequences $\{a_n\}$ and $\{b_n\}$, denote $A_n = \sum_{k=0}^n a_k$.
Then, for any $q \in \mathbb{N}$,

$$\sum_{n=0}^q a_n b_n = \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q = \sum_{n=0}^{q-1} \left(\sum_{k=0}^n a_k \right) \cdot (b_n - b_{n+1}) + b_q \cdot \sum_{n=0}^q a_n.$$

In particular, if $0 \leq p \leq q$,

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p.$$

Proof. Since $a_n = A_n - A_{n-1}$ (with $A_{-1} \stackrel{\text{def}}{=} 0$),

$$\begin{aligned} \sum_{n=0}^q a_n b_n &= \sum_{n=0}^q (A_n - A_{n-1}) b_n = \sum_{n=0}^q A_n b_n - \sum_{n=0}^q A_{n-1} b_n \\ &= \sum_{n=0}^q A_n (b_n - b_{n+1}) + \sum_{n=0}^q A_n b_{n+1} - \sum_{n=0}^q A_{n-1} b_n \\ &= \sum_{n=0}^q A_n (b_n - b_{n+1}) + A_q b_{q+1} \\ &= \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q. \end{aligned}$$

Particular,

$$\begin{aligned} \sum_{n=p}^q a_n b_n &= \sum_{n=0}^q a_n b_n - \sum_{n=0}^{p-1} a_n b_n \\ &= \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - \left(\sum_{n=0}^{p-2} A_n (b_n - b_{n+1}) + A_{p-1} b_{p-1} \right) \\ &= \sum_{n=p-1}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_{p-1} \\ &= \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_{p-1} b_{p-1} - A_{p-1} b_p + A_q b_q - A_{p-1} b_{p-1} \\ &= \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p. \end{aligned}$$

□

Note: Denote $A_n = \sum_{k=0}^n a_k$ and $\Delta b_n = b_{n+1} - b_n$, then

$$\sum_{n=0}^q a_n b_n = A_q b_q - \sum_{n=0}^{q-1} A_n \Delta b_n$$

It is similar to:

$$\int f \cdot g = Fg - \int Fg'$$

0.1.2 Abel's Test

Theorem 0.1.2. If $\sum a_n$ converges and $\{b_n\}$ is bounded and monotone sequence, then $\sum a_n b_n$ converges.

Proof. Since $\{b_n\}$ is bounded and monotone, it is convergent. Put $b = \lim_{n \rightarrow \infty} b_n$ and $A_n = \sum_{k=0}^n a_k$, $A_{-1} = 0$. WLOG, suppose $\{b_n\}$ is monotone decreasing. Then $\{b_n - b\}$ converges to zero and is decreasing.

Let $M > 0$ be a bound of $|A_n|$. Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ such that $n \geq N \implies |b_n - b| < \frac{\varepsilon}{2M}$.

By Abel's formula, for all $q \geq p \geq N$,

$$\begin{aligned} \left| \sum_{n=p}^q a_n (b_n - b) \right| &= \left| \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q (b_q - b) - A_{p-1} (b_p - b) \right| \\ &\leq M \cdot \left[\sum_{n=p}^{q-1} \underbrace{|b_n - b_{n+1}|}_{\geq 0} + \underbrace{|b_q - b|}_{\geq 0} + \underbrace{|b_p - b|}_{\geq 0} \right] \\ &= M \cdot \left[\sum_{n=p}^{q-1} (b_n - b_{n+1}) + (b_q - b) + (b_p - b) \right] \\ &= M \cdot [(b_p - b_q) + (b_q - b) + (b_p - b)] \\ &= 2M(b_p - b) < 2M \cdot \frac{\varepsilon}{2M} = \varepsilon \end{aligned}$$

Therefore, by the Cauchy Criterion for series, $\sum a_n (b_n - b)$ converges.

Moreover, since $\sum a_n$ converges, so is $b \cdot \sum a_n$. Finally,

$$\sum_{n=0}^{\infty} a_n (b_n - b) + \sum_{n=0}^{\infty} a_n \cdot b = \sum_{n=0}^{\infty} a_n b_n$$

converges as a sum of two convergent series. □

Note: If $b = 0$, it is sufficient for the convergence of $\sum a_n b_n$ that $\sum a_n$ be bounded.

In this case, $\sum a_n$ does not necessarily have to converge (it is called Dirichlet's Test).